

Unit 13, Lesson 4a: Weighted Averages of Directed Line Segments



Benchmark

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.



Learning Target

- ☐ I can determine the weighted average of two or more points on a directed line segment.
- ☐ I can explain how changing the weight moves the weighted average of a point on a directed line segment.

Teacher Notes: This supplemental lesson can be a companion to Unit 13, Lessons 4 and 5, or it can be used independently. If this is used independently, it will likely need to be implemented over two class periods. Additional components were included in this lesson to meet the request of teachers looking for additional resources for this benchmark, so this lesson was purposefully made longer to include additional components for both building conceptual understanding and practice. Note that the process of finding the weighted average on a number line and finding the weighted average on a line segment on the coordinate plane is the same. Teachers may also choose to use different components of Lessons 4, 4a, and 5 as they see fit for their classroom. This lesson scaffolds a student's understanding of the formula that can be used to determine points on a directed line segment that have a given weighted average. Teachers are encouraged to use correct language when discussing the line segments because the direction can affect the location of a point.

Teacher Notes: The warm-up helps students understand that concept of directed line segments by using rays. Teachers may also wish to relate the rays to vectors.

If time is an issue, teachers may wish to assign any combinations of sections 13.4a.1, 13.4a.2, and 13.4a.3 for homework prior to the day of the lesson. Teachers can then have students work with partners or groups to verify their answers and to edit their responses. If implementing in this way, teachers should circulate during the discussion or ask students for questions before the guided lesson.



13.4a.1 Warm-Up: Exploring Rays

Four rays are shown.

- 1. What do you notice about the four rays?

Answers will vary. Students may note the directions of the arrows, the length of the line segments, or the endpoints of the rays.

Each ray includes a line segment. These line segments are known as *directed line segments*. When naming a directed line segment, the first letter is the endpoint, and the second letter indicates the direction.

- 2. Which rays include the directed line segment HD ?

Rays 1 and 2

- 3. Which rays include the directed line segment DH ?

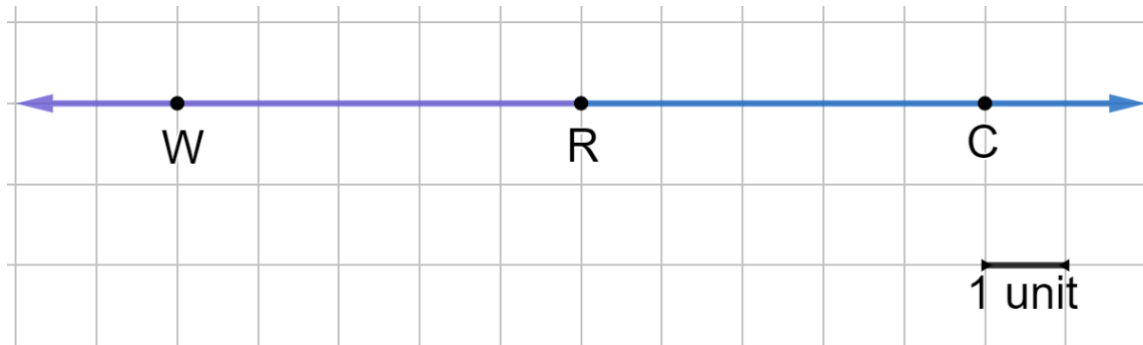
Rays 3 and 4

Ray 1	
Ray 2	
Ray 3	
Ray 4	



13.4a.2 Collaboration: Parts of Directed Line Segment

Rays RW and RC are shown on a grid, where each grid line represents 1 unit.



1. Complete the table for each

Ray	Directed Line Segment	Direction (Left or Right)	Length of the Directed Line Segment
$\overrightarrow{RW}$	RW	Left	5
$\overrightarrow{WC}$	WC	Right	10
$\overrightarrow{RC}$	RC	Right	5
$\overrightarrow{CW}$	CW	Left	10

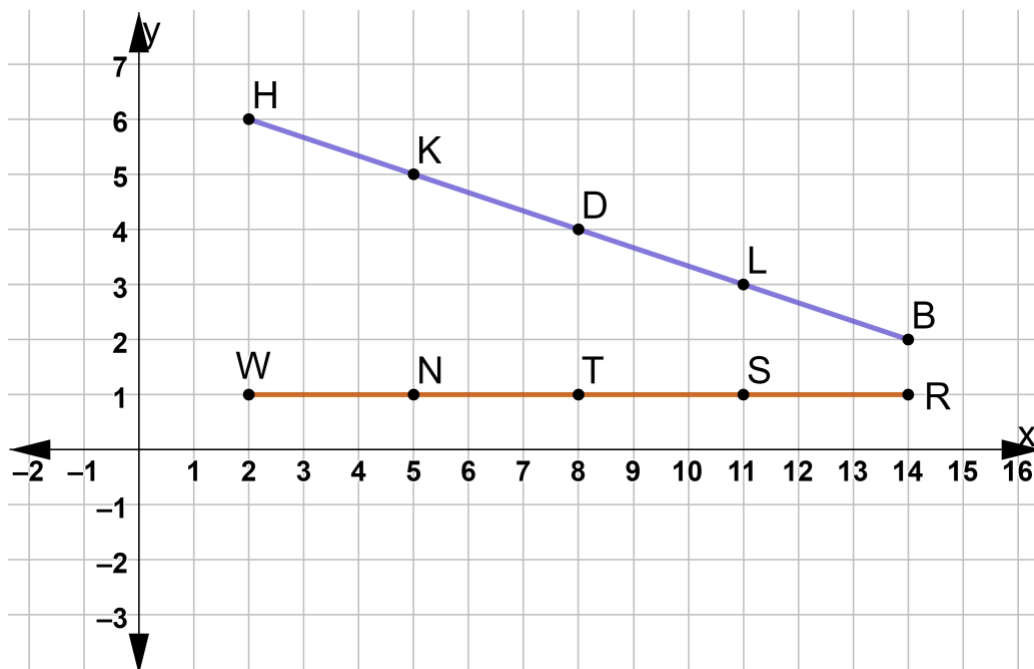
2. What is the ratio of the length of the directed line segment RW to the length of directed line segment RC ?

5 : 5 or 1 : 1

3. Complete the statement.

Point R is halfway between point W and point C .

Directed line segment HB and directed line segment WR are shown on the coordinate grid.



4. Complete each statement.
 - a. Directed line segment HB is divided into 4 parts.
 - b. Directed line segment WR is divided into 4 parts.
5. Point N divides directed line segment WR into the ratio of $\frac{WN}{NR}$. Point K divides directed line segment HB into the ratio of $\frac{KH}{BK}$. Discuss with your partner why these ratios have the same value. Summarize your discussion.

Answers may vary: Students may determine that the ratios are the same because the horizontal distance for both segments is the same and each line segment is divided into four parts. Both WN and KH are one of the four parts and NR and BK are three of the four parts.

6. Complete the statement.

The ratios $\frac{WN}{NR}$ and $\frac{KH}{BK}$ are
☐ part : part
☐ part : whole
 ratios.

Teacher Note: Teachers may want to ensure that students recall what meaning of part to part and part to whole.

7. Point K is closer to point H , so the ratio 1:3 is known as a *weighted average*. When a point does not divide a line segment at its midpoint, the result is that the line segment is divided so that one of the endpoints has more weight.

- a. On directed line segment HB , which endpoint is closer to point K ?

Point H

Point K divides directed line segment HB so that $HK : KB$ is 1 : 3. The formula $\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } B)$ can be used to determine the location of point K .

- b. Discuss with your partner why the ratio is 1 : 3 but the formula uses a denominator of 4. Summarize your discussion.

Answers will vary. Teachers should allow students to have productive struggle while answer this question. The lesson is scaffolded so that students will eventually understand this concept so allow students to have incorrect answers. Allow students to edit their answer at any point during the lesson.

8. Use the x -values to determine $\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } B)$.

$$\frac{3}{4}(2) + \frac{1}{4}(14) = \frac{6}{4} + \frac{14}{4} = \frac{20}{4} = 5$$

9. Use the x -values to determine $\frac{3}{4}(\text{point } W) + \frac{1}{4}(\text{point } R)$.

$$\frac{3}{4}(2) + \frac{1}{4}(14) = \frac{6}{4} + \frac{14}{4} = \frac{20}{4} = 5$$

10. What do you notice about the results of the two sums?

The sums are the same.

11. Point S divides directed line segment WR so that $WS : SR$ is 3 : 1. Complete the fractions to write the formula for the weighted average that would partition directed line segment WR resulting in point S .

$$\frac{1}{4}(\text{point } W) + \frac{3}{4}(\text{point } R)$$

Teacher Note: As you circulate, listen for students who only use the previous answers to be able to determine the relationship as opposed to students who use the part to part or part to whole ratios. Then, during the lesson empathize the relationship between the formula and the part to part & part to whole ratios

12. Verify your formula with your partner and make any necessary changes. Then, use your formula to show that it results in the x -value of point S .

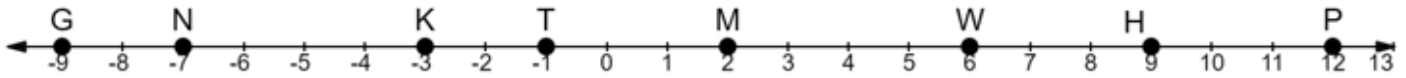
Student work should be shown here.

Teacher Note: Teachers should allow students the time to struggle on this. Gaining an understanding of the formula used in weighted averages will help students move from a part to part ratio or a part to whole ratio to the formula. While circulating, listen for confusion or misconceptions that can be addressed during Guided Instruction.



13.4a.3 Exploration: Weighted Averages on a Number Line

A number line is shown with several points plotted.



1. Complete the statements.
 - a. Directed line segment WG has a length of 15 units.
 - b. Directed line segment KP has a length of 15 units.
 - c. The ratio $\frac{WK}{WG}$ is $\frac{9}{15}$ or $\frac{3}{5}$.
 - d. The ratio $\frac{KW}{KP}$ is $\frac{9}{15}$ or $\frac{3}{5}$.
2. Discuss with your partner whether point K on directed line segment WG has the same weighted average as point W on directed line segment KP . Summarize your discussion.

Answers will vary. The answer is yes but how students express this may vary.

Teacher Note: As you circulate, listen for a misconception that shows that a student does not understand what a directed line segment is.

The ratio of $WK : KG$ can be written as $9 : 6$ or $3 : 2$. This is the weighted average. If you count 9 spaces from point W in the direction of point G , the result is point K . And if you go 6 spaces from point K in the direction of point G is point G .

Although it is easy to count in this situation it is not always easy to count spaces because a point can be **any** real number on a number line.

3. Complete the statements.
 - a. The ratio $\frac{WK}{WG}$ is a

☐ part : part
☒ **part : whole**

 ratio.
 - b. The ratio $\frac{WK}{KG}$ is a

☒ **part : part**
☐ part : whole

 ratio.

Point W partitions directed line segment KP so that point P weighs more than point K . This means that point W is closer to point P because point P is “heavier” than point K .

4. What is the part:whole ratio that is the result of point W partitioning directed line segment KP ?

9:15 or 3:5

5. What is the part:part ratio that is the result of point W partitioning directed line segment KP ?

9:6 or 3:2

6. Discuss with your partner how the part:whole ratio and the part:part ratio could be used to determine the formula that would give the location of Point W on the number line. Then, complete the formula.

$$\frac{6}{15}(\text{point } K) + \frac{9}{15}(\text{point } P) \text{ or } \frac{2}{5}(\text{point } K) + \frac{3}{5}(\text{point } P)$$

Teacher Note: As you circulate, listen for students who are struggling with the formula. Ask each group what the most difficult part of relating the ratios to the formula is and address these during Guided Instruction. Also, look for students who struggle using the equivalent ratios of 3 : 5 or 3 : 2.

7. Complete the table by determining the endpoint of each directed line segment that would be “heavier” given each point that partitions the segment.

Directed Line Segment	Point That Partitions the Segment	Endpoint That Is Heaviest
$\overrightarrow{GM}$	T	M
$\overrightarrow{MG}$	T	M
$\overrightarrow{TH}$	W	H
$\overrightarrow{PN}$	M	N
$\overrightarrow{PW}$	H	neither

Teacher Note: As you circulate, listen for students who may be struggle with the heaviest endpoint for $\overrightarrow{PW}$. Although students have learned about midpoints, relating them to weighted line segments may be a struggle.

8. Point T partitions directed line segment GM .

Directed Line Segment	Point That Partitions the Segment	Ratio	Numerical Ratio	Formula
$\overline{GM}$	T	$GT : TM$	$8 : 3$	$\frac{3}{11}(\text{point } G) + \frac{8}{11}(\text{point } M)$

- a. Discuss with your partner how the formula in the table relates to the numerical ratio and the heaviest endpoint. Summarize your discussion.

Answers may vary. Because point T is closer to point M on directed line segment GM then point M will have more weight than point G . Since the ratio $8:3$ is a part to part ratio, there are 11 whole parts and point M has the most weight.

- b. Use the values of point G and point M in the formula $\frac{3}{11}(\text{point } G) + \frac{8}{11}(\text{point } M)$ to verify that the result is point T .

$$\frac{3}{11}(-9) + \frac{8}{11}(2) = \frac{-27}{11} + \frac{16}{11} = \frac{-11}{11} = -1$$

9. Complete the table.

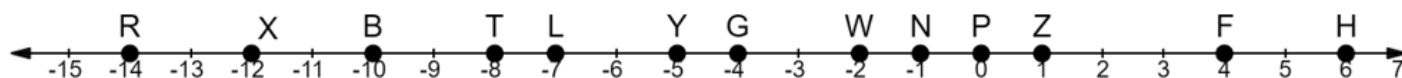
Directed Line Segment	Point That Partitions the Segment	Ratio	Numerical Ratio	Formula
$\overline{MG}$	T	$MT : TG$	$3 : 8$	$\frac{8}{11}(\text{point } M) + \frac{3}{11}(\text{point } G)$
$\overline{TH}$	W	$TW : WH$	$7 : 3$	$\frac{3}{10}(\text{point } T) + \frac{7}{10}(\text{point } H)$
$\overline{PN}$	M	$PM : MN$	$10 : 9$	$\frac{9}{19}(\text{point } P) + \frac{10}{19}(\text{point } N)$
$\overline{PW}$	H	$PH : HW$	$3 : 3$ or $1 : 1$	$\frac{3}{6}(\text{point } P) + \frac{3}{6}(\text{point } W)$ or $\frac{1}{2}(\text{point } P) + \frac{1}{2}(\text{point } W)$



13.4a.4 Guided Instruction: Weighted Averages on a Number Line

Teacher Notes: This component builds from student discussions during the previous 3 components. It progresses from providing a number line and using integers to no number line and using real numbers in various formats. Teachers are encouraged to allow students to use calculators.

1. A number line is shown with several points labeled.



- a. Determine the ratio of $HG : GB$.

10:6 or 5:3

Teachers should provide both answers during instruction.

- b. For the directed line segment HB , which endpoint is heaviest in the ratio $HG : GB$?

Point B

- c. Write the formula for the directed line segment HB that would result in point G .

$$\frac{6}{16}(\text{point H}) + \frac{10}{16}(\text{point B}) = \text{point G} \text{ or } \frac{3}{8}(\text{point H}) + \frac{5}{8}(\text{point B}) = \text{point G}$$

Teachers should provide both answers during instruction.

- d. Which point on directed line segment HB and directed line segment BH would have the same formula?

Point W

- e. Which point on directed line segment GR partitions the line segment into line segments in a ratio of 4 : 1?

Point X

- f. For the directed line segment GR , which endpoint is heaviest in the ratio 2 : 3?

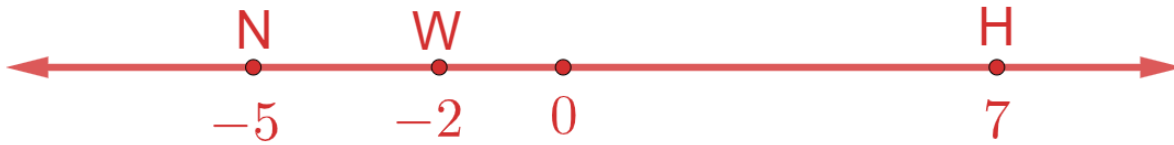
Point G

- g. Write the formula for the directed line segment GR that would partition the line segment in a ratio of 7 : 3.

$$\frac{3}{10}(\text{point G}) + \frac{7}{10}(\text{point R})$$

2. Points H , N and W are on a number line. Point H is located at 7. Point W is located at -2 . Point W partitions directed line segment NH in the ratio $1 : 3$.

- a. Draw a number line that shows points H and W are on a number line.



- b. Use the number line in part a to show how point N could be determined.
Answers will vary. Students may use the distance from HW and solve the proportion $\frac{1}{3} = \frac{x}{9}$ to determine the distance N is from W .
- c. For the ratio $1 : 3$, which endpoint of the directed line segment NH is the heaviest?
Point N
- d. Use the formula for partitioning line segments to write an equation that can be used to determine the location of endpoint N of directed line segment NH .
 $\frac{3}{4}(\text{point } N) + \frac{1}{4}(\text{point } H) = \text{point } W$
- e. Use the equation to determine the location of endpoint N of directed line segment NH .
 $\frac{3}{4}(\text{point } N) + \frac{1}{4}(7) = -2 \quad \frac{3}{4}(\text{point } N) + \frac{7}{4} = -2 \quad \frac{3}{4}(\text{point } N) = -\frac{15}{4} \quad \text{point } N = -5$

3. Point K partitions directed line segment RD in the ratio $3 : 5$. On a number line, point R is located at -23 and point K is located at 1. Determine the location of point D .

$$\frac{5}{8}(\text{point } R) + \frac{3}{8}(\text{point } D) = \text{point } K \quad \frac{5}{8}(-23) + \frac{3}{8}(\text{point } D) = 1 \quad \frac{-115}{8} + \frac{3}{8}(\text{point } D) = 1$$

$$\frac{3}{8}(\text{point } D) = \frac{123}{8} \quad \text{point } D = 41$$

Teachers should allow students to share their methods of determining point D . Students may use proportions or the number line to find the location of point D .

4. Point Y partitions directed line segment HT in the ratio $1 : 1$. On a number line, point T is located at $-4\frac{3}{4}$ and point Y is located at $-8\frac{1}{8}$. Determine the location of point H .

$$\frac{1}{2}(\text{point } H) + \frac{1}{2}(\text{point } T) = \text{point } Y \quad \frac{1}{2}(\text{point } H) + \frac{1}{2}\left(-4\frac{3}{4}\right) = -8\frac{1}{8} \quad \frac{1}{2}(\text{point } H) + \frac{1}{2}\left(-\frac{19}{4}\right) = -\frac{65}{8}$$

$$\frac{1}{2}(\text{point } H) + -\frac{19}{8} = -\frac{65}{8} \quad \frac{1}{2}(\text{point } H) = -\frac{46}{8} \quad \text{point } H = -\frac{46}{4} \text{ or } -\frac{23}{2}$$

Teachers should allow students to share their methods of determining point H . Students may use proportions or the number line to find the location of point H . Also, students may recognize a $1 : 1$ ratio as determining the midpoint and use that information to determine the location of point H .

5. Point N partitions directed line segment KW in the ratio $9 : 16$. On a number line, point N is located at $-8\frac{2}{3}$ and point W is located at $55\frac{1}{3}$. Determine the location of point K .

$$\frac{16}{25}(\text{point } K) + \frac{9}{25}(\text{point } W) = \text{point } N \quad \frac{16}{25}(\text{point } K) + \frac{9}{25}\left(55\frac{1}{3}\right) = -8\frac{2}{3} \quad \text{point } K = -\frac{134}{3}$$



A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

6. Discuss with your partner how to determine from a ratio if a point partitions a directed line segment equally. Summarize your discussion.

Answers will vary.

Teacher Note: *Lead students to understand how a midpoint is represented using ratios and why the weighted average is $1 : 1$.*



13.1a.5 Guided Practice: Weighted Averages

1. Point B partitions directed line segment YN in the ratio $7 : 13$. On a number line, point N is located at 17.4 and point B is located at 38.4 . Determine the location of point Y .
 77.4
2. Point W partitions directed line segment NB . On a number line, point B is located at $-22\frac{3}{4}$, point N is located at $8\frac{1}{4}$, and point W is located at $-\frac{7}{4}$. Determine the weighted average.
 $40 : 84$ or any equivalent



13.4a.6 Your Turn!

1. Point L partitions directed line segment TS . On a number line, point S is located at -72 , point L is located at -32 , and point T is located at 43 . Determine the weighted average.
 $75 : 40$ or any equivalent
2. Point R partitions directed line segment TK in the ratio $1 : 1$. On a number line, point T is located at $-\frac{49}{2}$ and point R is located at $-\frac{45}{8}$. Determine the location of point K .
 $\frac{106}{8}$ or any equivalent



13.4a.7 Wrap-Up: Ticket Out the Door

1. Point B partitions directed line segment LF in the ratio $5 : 7$. On a number line, point F is located at 13.62 and point B is located at -3.88 . Determine the location of point L .
 -16.38